

1. The number of ways of giving 20 distinct oranges to 3 children such that each child gets at least one orange is _____.
[2023 (06 Apr Shift 1)]
2. All the letters of the word PUBLIC are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word PUBLIC is
[2023 (06 Apr Shift 2)]
(1) 576
(2) 578
(3) 580
(4) 582
3. The number of 4-letter words, with or without meaning, each consisting of 2 vowels and 2 consonants, which can be formed from the letters of the word UNIVERSE without repetition is _____.
[2023 (06 Apr Shift 2)]
4. The number of arrangements of the letters of the word "INDEPENDENCE" in which all the vowels always occur together is
[2023 (08 Apr Shift 1)]
(1) 16800
(2) 33600
(3) 18000
(4) 14800
5. The number of ways, in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together is
[2023 (08 Apr Shift 1)]
(1) 720
(2) $126(5!)^2$
(3) $7(360)^2$
(4) $7(720)^2$
6. If the number of words, with or without meaning, which can be made using all the letters of the word MATHEMATICS in which C and S do not come together, is $(6!)^k$ then k is equal to
[2023 (08 Apr Shift 2)]
(1) 2835
(2) 5670
(3) 1890
(4) 945
7. The number of permutations, of the digits 1, 2, 3, ..., 7 without repetition, which neither contain the string 153 nor the string 2467, is _____.
[2023 (10 Apr Shift 1)]
8. Some couples participated in a mixed doubles badminton tournament. If the number of matches played, so that no couple played in a match, is 840, then the total numbers of persons, who participated in the tournament, is _____.
[2023 (10 Apr Shift 1)]
9. Eight persons are to be transported from city A to city B in three cars of different makes. If each car can accommodate at most three persons, then the number of ways, in which they can be transported, is
[2023 (10 Apr Shift 2)]
(1) 1120
(2) 3360
(3) 1680
(4) 560
10. The sum of all the four-digit numbers that can be formed using all the digits 2, 1, 2, 3 is equal to _____.
[2023 (10 Apr Shift 2)]
11. The number of triplets (x, y, z) where x, y, z are distinct non negative integers satisfying $x + y + z = 15$, is
[2023 (11 Apr Shift 1)]
(1) 80
(2) 136
(3) 114
(4) 92
12. In an examination, 5 students have been allotted their seats as per their roll numbers. The number of ways, in which none of the students sits on the allotted seat, is _____.
[2023 (11 Apr Shift 1)]

13. If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers, then the serial number of the word THAMS is
[2023 (11 Apr Shift 2)]
(1) 103
(2) 102
(3) 101
(4) 104
14. The number of five-digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to
[2023 (12 Apr Shift 1)]
(1) 132
(2) 120
(3) 72
(4) 96
15. Let the digits a, b, c be in A.P. Nine-digit numbers are to be formed using each of these three digits thrice such that three consecutive digits are in A.P. at least once. How many such numbers can be formed?
[2023 (12 Apr Shift 1)]
16. The number of seven digit positive integers formed using the digits 1, 2, 3 and 4 only and sum of the digits equal to 12 is _____.
[2023 (13 Apr Shift 1)]
17. All words, with or without meaning, are made using all the letters of the word *MONDAY*. These words are written as in a dictionary with serial numbers. The serial number of the word *MONDAY* is
[2023 (13 Apr Shift 2)]
(1) 327
(2) 328
(3) 324
(4) 326
18. Total numbers of 3-digit numbers that are divisible by 6 and can be formed by using the digits 1, 2, 3, 4, 5 with repetition, is _____.
[2023 (13 Apr Shift 2)]
19. The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8, if repetition of digits is allowed, is
[2023 (15 Apr Shift 1)]
(1) 21
(2) 20
(3) 22
(4) 18
20. A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is _____.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

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|----------|-------------|----------|----------|---------|---------|------------|-----------|
| 1. (171) | 2. (4) | 3. (432) | 4. (1) | 5. (2) | 6. (2) | 7. (4898) | 8. (16) |
| 9. (3) | 10. (26664) | 11. (3) | 12. (44) | 13. (1) | 14. (2) | 15. (1260) | 16. (413) |
| 17. (1) | 18. (16) | 19. (3) | 20. (72) | | | | |
1. (171)

Given,

We have to distribute 20 identical oranges to 3 children,

Let children A get O_A oranges, children B gets O_B oranges and children C gets O_C oranges,

So, $O_A + O_B + O_C = 20$

Now this can be distributed using the multinomial theorem,

So, Number of ways will be,

$$= \text{coefficient of } x^{20} \text{ in } (x + x^2 + \dots + x^{18})^3$$

$$= \text{coefficient of } x^{20} \text{ in } x^3(1 + x + x^2 + \dots + x^{17})^3$$

$$= \text{coefficient of } x^{17} \text{ in } \left(\frac{1-x^{18}}{1-x}\right)^3$$

$$= \text{coefficient of } x^{17} \text{ in } (1-x)^{-3}$$

$$= {}^{19}C_2 = 171$$

As we know that expansion of $(1-x)^{-3}$ is $1 + {}^3C_1x + {}^4C_2x^2 + {}^5C_3x^3 + \dots + {}^{19}C_{17}x^{17} + \dots \infty$

Note: This is bonus question as here 20 oranges are considered identical but in question it is considered distinct.

2. (4)

The given word is PUBLIC

Arranging the letters alphabetically, we get

BCILPU

When the word starts with any of the letters B/C/I/L, the number of possibilities = $5! \times 4 = 480$

Now when the word starts with PB, then the number of possibilities = $4! = 24$

Now when the word starts with PC, then the number of possibilities = $4! = 24$

Now when the word starts with PI, then the number of possibilities = $4! = 24$

Now when the word starts with PL, then the number of possibilities = $4! = 24$

Now when the word starts with PUBC, then the number of possibilities = $2! = 2$

Now when the word starts with PUBI, then the number of possibilities = $2! = 2$

Now when the word starts with PUBLIC, then the number of possibilities = 1

Now when the word starts with PUBLIC, then the number of possibilities = 1

$$\text{Rank} = 480 + 24 \times 4 + 2 \times 2 + 1 \times 2 = 582$$

Hence, rank of the word PUBLIC is 582.

3. (432)

Given,

We have to make four letter word from UNIVERSE each consisting 2 vowels and 2 consonants,

Now in UNIVERSE, we have,

E, E, I, U, (Vowels) + N, R, S, V (Consonants)

2 different vowels + 2 consonants can be arrange in ${}^3C_2 \cdot {}^4C_2 \cdot 4! = 3 \times 6 \times 24 = 432$ ways

{ 3C_2 is taken because of repetition of letter E }

4. (1)

There are 5 vowels in the given word which are 4E's & 1I.

Since they have to always occur together we take them as a single object $E E E E I$ for the time being.

This single object together with 7 remaining object will account for 8 objects.

There 8 objects in which there are 3N's & 2D's can be arrangement in $\frac{8!}{3!2!}$ ways.

Corresponding to each of their arrangements the 5 vowels E, E, E, E & I which can be arranged in $\frac{5!}{4!}$

Hence, required number of arrangements.

$$= \frac{8!}{3!2!} \times \frac{5!}{4!} = 16800$$

Hence this is the correct option.

5. (2)

Given,

7 boys and 5 girls are to be seated around a circular such that no two girls to be seated together,

Now we know that n objects can be arranged in a circle in $(n - 1)!$ ways.

Let us first arrange 7 boys in circular arrangement in $(7 - 1)!$ ways.

Now there will be 7 gaps.

So let us select any 5 gaps out of 7 gaps and arrange 5 girls in the chosen gaps. This can be done in ${}^7C_5 \times 5!$ ways.

Hence, required arrangements are $6! \times {}^7C_5 \times 5!$

$$= 6 \times 5! \times \frac{7 \times 6}{2} \times 5!$$

$$= 126(5!)^2.$$

Therefore, required arrangements are $126(5!)^2$

6. (2)

The word 'MATHEMATICS' consists of 11 letters including two M 's, two T 's and two A 's,

Now, first we will arrange letters $\underline{M} \underline{A} \underline{T} \underline{H} \underline{E} \underline{M} \underline{A} \underline{T} \underline{I}$.

Here, we have 10 gaps to fill C & S .

The number of words with or without meaning can be formed from the word MATHEMATICS when C , S does not come together is

$$= \frac{9!}{2! \cdot 2! \cdot 2!} \times {}^{10}C_2 \times 2!$$

$$= \frac{9 \times 8 \times 7 \times 6!}{2! \cdot 2!} \times \frac{10 \times 9}{2}$$

$$= 5670$$

Hence this is the correct option.

7. (4898)

Given,

The number $1, 2, 3, \dots, 7$

Now total numbers of 7 digit number will be $7!$ without repetition,

Now total numbers which contain string 154 will be $5!$

Total numbers which contain string 2367 will be $4!$

And number which contain both string 2367 & 154 will be 2

So, total number which does not contain string 154 or 2367 will be $7! - (5! + 4! - 2)$

$$= 5040 - (120 + 24 - 2)$$

$$= 5040 - 142 = 4898$$

Hence this is the required answer.

8. (16)

Given,

Some couples participated in a mixed doubles badminton tournament. If the number of matches played, so that no couple played in a match, is 840,

Now Let total number of persons be $2n$, so number of couples will be n

Now number of matches played, so that no couple played in a match is given by,

$$\Rightarrow {}^nC_2 \cdot {}^{n-2}C_2 \cdot 2 = 840$$

{as if we choose couple as $\{A, B\}$ and other couple as $\{C, D\}$ and taking other two couple as $\{x, y\}$ & $\{w, z\}$, so the possible combination of the match where no couple played in a match will be, $\{A, x\} \rightarrow \{C, w\}$ & $\{C, x\} \rightarrow \{A, w\}$, hence there are two possible combination after choosing }

$$\Rightarrow n(n-1)(n-2)(n-3) = 5 \cdot 6 \cdot 7 \cdot 8$$

So, $n = 8$

Hence, the number of persons will be, $2n = 16$

9. (3)

Let us find the different ways so that 8 persons can travel in 3 cars.

C_1	C_2	C_3
3	3	2
2	3	3
3	2	3

Hence we have 3 ways.

Now the number of ways to distribute 8 persons such that they can travel in 3 cars with any car carrying maximum of 3 persons is $\left(\frac{8!}{3!3!2!}\right) \times 3$
= 1680

Therefore, the required answer is 1680

10. (26664)

Total 4 digit numbers using the digits 1, 2, 2, 3 is

$$= \frac{4!}{2!} = 12$$

Numbers are

$\begin{array}{r} _ _ _ 3 \\ _ _ _ 3 \\ _ _ _ 3 \\ _ _ _ 2 \\ _ _ _ 2 \\ _ _ _ 2 \\ _ _ _ 2 \\ _ _ _ 2 \\ _ _ _ 2 \\ _ _ _ 1 \\ _ _ _ 1 \\ _ _ _ 1 \end{array}$

Required sum is

$$\begin{aligned}
 &= \underbrace{[3 \times 3 + 2 \times 6 + 1 \times 3]}_{\text{unit digit's sum}} + \underbrace{[(3 \times 3 + 2 \times 6 + 1 \times 3) \times 10]}_{\text{ten's place sum}} + \underbrace{[(3 \times 3 + 2 \times 6 + 1 \times 3) \times 100]}_{\text{hundred's place sum}} + \underbrace{[(3 \times 3 + 2 \times 6 + 1 \times 3) \times 1000]}_{\text{thousand's place sum}} \\
 &= [24] + [24 \times 10] + [24 \times 100] + [24 \times 1000] \\
 &= 24 + 240 + 2400 + 24000 = 26664
 \end{aligned}$$

11. (3)

Given,

$$x + y + z = 15$$

Now we know that,

Non-negative integral solution of equation $a + b + c = n$ is given by ${}^{n+3-1}C_{3-1}$ where $a = b = c$ & $a = b \neq c$ are also possibilities

So by above formula we get,

Total number of non-negative solution will be, ${}^{15+3-1}C_{3-1} = {}^{17}C_2$

Now solving If any of these 2 are equal

So, the equation will become $x + 2y = 15$

Now finding possible cases we get,

$$y = 0 \quad x = 15$$

$$y = 1 \quad x = 13$$

$$y = 2 \quad x = 11$$

$$y = 3 \quad x = 9$$

$$y = 4 \quad x = 7$$

$$y = 5 \quad x = 5 \rightarrow x = y = z = 5$$

$$y = 6 \quad x = 3$$

$$y = 7 \quad x = 1$$

So, total possibilities where x, y & z are distinct will be,

$$= {}^{17}C_2 - {}^3C_2 \times 8 + 2$$

{Note adding 2 because the cases $x = y = z$ is subtracted three times}

$$= 136 - 24 + 2 = 114$$

12. (44)

5 boys with allotted roll numbers and seat numbers are seated in such a way that no one sits on the allotted seats. The number of ways of such seating arrangement is equal to the derangement of 5 boys i.e.,

$$= 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right)$$

$$= 5! - 5! + 60 - 20 + 5 - 1$$

$$= 44$$

13. (1)

The given word is MATHS

Arranging the letters alphabetically, we get

AHMST

When the word starts with any of the letters A/H/M/S, the number of possibilities = $4! \times 4 = 96$

Now when the word starts with TA, then the number of possibilities = $3! = 6$

Now when the word starts with THAMS, then the number of possibilities = 1

$$\text{Rank} = 96 + 6 + 1 = 103$$

Hence, rank of the word THAMS is 103.

14. (2)

We need to find the number of 5 digit numbers greater than 40000 and divisible by 5.

Case I : Numbers start with 5.

Here if the number starts with 5 it should end with 0 to be divisible by 5.

$$\begin{array}{c} 5 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} \begin{array}{c} 0 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} = 4 \times 3 \times 2 = 24$$

Case II : Number starts with 7 and the last digit can have 0 or 5.

$$\begin{array}{c} 7 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} \begin{array}{c} 0 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} = 48$$

Case III : The number starts with 9 and the last digit should have 0 or 5.

$$\begin{array}{c} 9 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} \begin{array}{c} 0 \\ \downarrow \\ \downarrow \\ \downarrow \end{array} = 48$$

Total numbers = 120.

Hence this is the correct option.

15. (1260)

Three numbers a, b, c are in A.P. and they are used to making 9-digit number using each digit thrice such that atleast three consecutive digits are in A.P., So, c, b, a will also be in A.P.

Now, we have 7 position to put (a, b, c) or (c, b, a) as $\overbrace{\quad \quad \quad}^{7 \text{ positions}}$

Number of such numbers is

$$= {}^2C_1 \cdot {}^7C_1 \cdot \frac{6!}{2! \cdot 2! \cdot 2!}$$

$$= 1260$$

16. (413)

The given numbers are 1, 2, 3, 4 and sum of digits should be 12.

Case 1 : $\underline{4} \ \underline{3} \ \underline{1} \ \underline{1} \ \underline{1} \ \underline{1} \ \underline{1}$, Here the sum of the digits is 12.

The number of ways arranging them is $= \frac{7!}{5!} = 42$.

Case 2 : $\underline{4} \ \underline{2} \ \underline{2} \ \underline{1} \ \underline{1} \ \underline{1} \ \underline{1}$.

The number of ways arranging them is $= \frac{7!}{2!4!} = 105$

Case 3 : $\underline{3} \ \underline{2} \ \underline{2} \ \underline{2} \ \underline{1} \ \underline{1} \ \underline{1}$.

The number of ways arranging them is $= \frac{7!}{3!3!} = 140$

Case 4 : $\underline{3} \ \underline{3} \ \underline{2} \ \underline{1} \ \underline{1} \ \underline{1} \ \underline{1}$.

The number of ways arranging them is $= \frac{7!}{2!4!} = 105$

Case 5 : $\underline{2} \ \underline{2} \ \underline{2} \ \underline{2} \ \underline{2} \ \underline{1} \ \underline{1}$

The number of ways arranging them is $= \frac{7!}{5!2!} = 21$

The total number of ways are $= 42 + 105 + 140 + 105 + 21$

$= 413$

Therefore, the required answer is 413.

17. (1)

The given word is MONDAY

Arranging the letters alphabetically, we get
ADMNOY

When the word starts with any of the letters A/D, the number of possibilities = $5! \times 2 = 240$

Now when the word starts with MA, then the number of possibilities = $4! = 24$

Now when the word starts with MD, then the number of possibilities = $4! = 24$

Now when the word starts with MN, then the number of possibilities = $4! = 24$

Now when the word starts with MOA, then the number of possibilities = $3! = 6$

Now when the word starts with MOD, then the number of possibilities = $3! = 6$

Now when the word starts with MONA, then the number of possibilities = $2! = 2$

Now when the word starts with MONDAY, then the number of possibilities = 1

Rank = $240 + 24 \times 3 + 6 \times 2 + 2 + 1 = 327$

Hence, rank of the word MONDAY is 327.

SHORTCUT METHOD

Another method is a shortcut, first number the alphabet and then check how many number are there on right hand side less than that number and then start number in factorial in decreasing order and then multiply and add them and in last add one to find rank,

3 5 4 2 1 6

M O N D A Y

2 3 2 1 0 0

5! 4! 3! 2! 1! 0!

$\therefore$ Rank = $(2 \times 5! + 3 \times 4! + 2 \times 3! + 1 \times 2!)$

= $240 + 72 + 12 + 2 + 1 = 327$

18. (16)

A number is divisible by 6 when it is divisible by 2 and 3 both.

The three digit number can be $\overline{ab2}$.

$a + b$ can be 4, 7, 10

If $a + b = 4$ then (a, b) can be (1, 3), (3, 1), (2, 2)

If $a + b = 7$ then (a, b) can be (2, 5), (5, 2), (3, 4), (4, 3)

If $a + b = 10$ then (a, b) can be (5, 5)

So, 8 such cases when 2 is at unit's place

Similarly, there exist 8 such cases when 4 is at unit's place.

Total = 16 cases.

Hence this is the required answer.

19. (3)

Given,

Three-digit number to be formed which are divisible by 3 using the digits 1, 3, 5, 8 and repetition is allowed,

Now taking case (1), where all digits are same, we get

(1, 1, 1), (3, 3, 3), (5, 5, 5), (8, 8, 8) $\rightarrow$ 4 ways

Now taking case (2), where 2-digit are same and one is distinct, we get

$(5, 5, 8) \rightarrow \frac{3!}{2!} = 3$ ways, $(8, 8, 5) \rightarrow 3$ ways

Now taking case (3), where all are distinct, we get

$(1, 3, 5) \rightarrow 6$ ways, $(1, 8, 3) \rightarrow 6$ ways

So, total ways will be $4 + 3 + 3 + 6 + 6 = 22$

20. (72)

Let the 4-digit pin be $abcd$

Now given sum of first two is equal to last two, so $a + b = c + d$

And number available are 0, 1, 2, 3, 4, 5, 6, 7

Now Let $a + b = c + d = \lambda$

Also the greatest digit that he has used is 7

So, $a + b = c + d = \lambda \geq 7$

Now taking all cases starting from $\lambda = 7$ we get,

$\lambda = 7$, $(a, b) \rightarrow (0, 7)$ or $(7, 0)$, then $(c, d) \rightarrow (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ and vice-versa, so $2 \times 12 = 24$ numbers

$\lambda = 8$, $(a, b) \rightarrow (1, 7)$ or $(7, 1)$, then $(c, d) \rightarrow (2, 6), (3, 5), (6, 2), (5, 3)$ and vice versa so total $2 \times 8 = 16$ numbers,

$\lambda = 9$, $(a, b) \rightarrow (2, 7)$ or $(7, 2)$ then $(c, d) \rightarrow (3, 6), (4, 5), (5, 4), (6, 3)$ and vice versa, so total $2 \times 8 = 16$ numbers,

$\lambda = 10$, $(a, b) \rightarrow (3, 7), (7, 3)$, so $(c, d) \rightarrow (6, 4), (4, 6) \rightarrow 8$ numbers

$\lambda = 11$, $(a, b) \rightarrow (4, 7), (7, 4)$, so $(c, d) \rightarrow (6, 5), (5, 6) \rightarrow 8$ numbers

$\lambda = 12, 13$ & 14 are not possible,

So, Total numbers $= 24 + 16 + 16 + 8 + 8 = 72$